



Hypersthene

△ Banded hypersthene rough with crystal striations

The name **hypersthene** is still retained as a gemstone variety, even though it has been largely abandoned as a mineral name (the mineral is now considered a middle member of a related series of silicates called pyroxenes). Hypersthene is commonly grey, brown, or green in colour. As a gem, it is noted for its copper-red iridescence, caused in part by inclusions of the minerals hematite and goethite. It is most frequently cut *en cabochon* because it can be too dark to facet. When faceted, stones are often somewhat cloudy, although intense in colour.

Specification

Chemical name Magnesium, iron silicate | **Formula** (Mg,Fe) (Si₂O₆) | **Colours** Grey, brown, green | **Structure** Orthorhombic
Hardness 5.5 | **SG** 3.35 | **RI** 1.65–1.67 | **Lustre** Vitreous
Streak White | **Locations** India, Germany, Norway, Greenland



Double-terminated crystal

Rare, high-quality crystal | Rough | Hypersthene occasionally forms well-developed crystals, such as this double-terminated crystal resting on a rock groundmass.



Striated surface

Hypersthene rock | Rough | Hypersthene in its rough form, as here, can resemble the mineral hornblende, and so the two are often confused. Hypersthene is the harder mineral.



Tumble polish | Cut | Like all gemstones, hypersthene comes in a number of grades. The lowest-grade material is often tumble polished, as in this example.

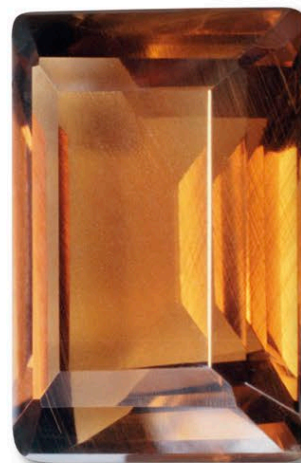
Once a gem, always a gem

Pyroxene or hypersthene?

As scientific instrumentation evolves, previously unknown chemical nuances in minerals are revealed. This can affect the naming conventions that apply to them, particularly in terms of the different requirements of science and commerce. For example, now that hypersthene is known to be a member of the mineral series pyroxene, it no longer technically needs a separate name. To the commercial market, however, “hypersthene” is a well-known gem name, and remains in use.



Scientific analysis Close examination of hypersthene has led to it being mineralogically reclassified.



Faceted hypersthene | Cut | This rare, step-cut hypersthene shows good colour and clarity, but characteristically lacks brilliance. The corners are rounded to prevent chipping.



Shaped cabochon | Carved | This leaf-like carving of the hypersthene variety bronzite glows from platy inclusions (flat, thin mineral crystals), giving it a subtle metallic sheen.